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AI-Powered Translation and Contextual Interpretation of Ancient Telugu Inscriptions Using Large Language Models

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

DSA 0304 – NATURAL LANGUAGE PROCESSING
to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

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COMPUTER SCIENCE AND ENGINEERING (AI&DS)

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**AI-Powered Translation and Contextual Interpretation of Ancient Telugu Inscriptions Using Large Language Models**” has been carried out by **B Akshaya (192424336)** under the supervision of **DR T Devi mam** and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Computer Science and Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **B Akshaya** of the **[AIDS]**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**[AI-Powered Translation and Contextual Interpretation of Ancient Telugu Inscriptions Using Large Language Models]**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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(Mandatory for all team projects)

Student Name	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution
B. Pallavi	Image Processing and Text Extraction	Developed image preprocessing, enhancement and OCR workflow	Tested image quality and text extraction	Module documentation and literature support	25%
A. Ashriktha Reddy	Translation Using LLMs	Developed text preprocessing, error correction and LLM-based translation	Tested translation accuracy and contextual relevance	Module design and implementation documentation	25%
B. Akshaya	Contextual Interpretation and Knowledge Extraction	Developed NER, information extraction, relation extraction and summarization workflow	Tested extracted entities, relationships and generated summaries	Module testing, results and analysis documentation	25%
Ch. Haritha	Digital Archive and Interactive Dashboard	Developed archive structure, search and dashboard integration	Tested data storage, search and dashboard functions	Integration and final project documentation	25%
Total					100%

ABSTRACT

Ancient Telugu inscriptions are valuable sources of historical, cultural, religious, and social information. They provide important details about rulers, temples, land grants, administration, traditions, and the development of the Telugu language. However, understanding these inscriptions is difficult because they often contain old Telugu scripts, archaic vocabulary, damaged characters, and grammatical structures that are different from modern Telugu. Manual translation and interpretation require expert knowledge of Telugu linguistics, history, and epigraphy and can be time-consuming. This project proposes an **AI-Powered Translation and Contextual Interpretation System for Ancient Telugu Inscriptions using Large Language Models (LLMs)**. The system accepts an image of an ancient Telugu inscription and applies image preprocessing and Optical Character Recognition (OCR) techniques to extract the written text. The extracted text is then processed using Natural Language Processing (NLP) techniques to improve readability, transliterate historical text, and convert it into modern Telugu and English. A Large Language Model is used to understand the linguistic and historical context of the extracted content and generate meaningful translations and explanations. The proposed system can provide the original extracted text, transliteration, modern Telugu translation, English translation, and contextual interpretation in a structured format. It can help identify important information such as names, places, rulers, dates, religious references, and administrative details when available in the inscription. The system aims to reduce the effort involved in initial inscription analysis and make ancient Telugu heritage more accessible to students, researchers, historians, and the general public. By combining OCR, NLP, and LLM technologies, the project provides a modern approach for the digital preservation, translation, and interpretation of ancient Telugu inscriptions while supporting further research into Telugu language and cultural heritage.

Keywords

Ancient Telugu Inscriptions, Artificial Intelligence (AI), Large Language Models (LLMs), Natural Language Processing (NLP), Optical Character Recognition (OCR), Telugu Language Processing, Machine Translation, Historical Text Analysis, Transliteration, Contextual Interpretation, Text Extraction, Digital Heritage Preservation, Historical Linguistics, Cultural Heritage, Ancient Script Recognition, Semantic Analysis, Language Understanding, Automated Translation, Knowledge Extraction, Epigraphy, Intelligent Interpretation System.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Ancient Telugu inscriptions are valuable sources of historical, cultural, social, political, and religious information. These inscriptions, commonly found on stone monuments, temple walls, pillars, copper plates, and other historical artifacts, provide important evidence about the rulers, dynasties, geographical locations, religious practices, donations, wars, administrative systems, and social conditions of earlier periods. However, understanding ancient inscriptions is a challenging task. Many inscriptions have been damaged over time because of environmental conditions, erosion, physical deterioration, and incomplete preservation. Even when the text is available in digital form, interpreting its historical meaning requires significant domain knowledge and manual effort. The information contained within an inscription is often unstructured, making it difficult to identify important entities and relationships automatically. Natural Language Processing (NLP) provides techniques for analyzing textual information and extracting meaningful knowledge from unstructured text. In this project, the contextual interpretation and knowledge extraction module focuses on analyzing translated inscription text and identifying important historical information.

1.2 Need for the Project

1. Difficulty in Understanding Historical Context

A direct translation of an ancient inscription may not provide complete historical meaning. Researchers need to understand who is being mentioned, where an event occurred, when it happened, and how different entities are connected. Therefore, contextual interpretation is necessary.

2. Unstructured Historical Information

Inscription text generally contains information in an unstructured form. Important details such as names of rulers, temples, places, dates, and events are embedded within sentences. Extracting this information manually from a large collection is time-consuming.

3. Manual Analysis Requires Expert Knowledge

Traditional historical analysis depends heavily on specialists. Manual identification of entities and relationships requires significant time and expertise, particularly when dealing with large numbers of inscriptions.

4. Need for Digital Preservation

Historical information should be converted into structured digital knowledge so that it can be preserved and accessed efficiently. Structured records make it easier to search inscriptions based on kings, dynasties, locations, temples, or historical periods.

5. Improved Accessibility

An automated contextual interpretation system can make historical information more accessible to researchers, students, historians, and the general public. Summaries and structured knowledge can help users understand the main content of inscriptions without analysing every sentence manually.

6. Engineering Significance

The project demonstrates the application of modern NLP and LLM technologies to a cultural heritage problem. It integrates multiple NLP tasks into a single knowledge extraction pipeline, transforming textual data into meaningful and structured historical information.

1.3 Problem Statement

Ancient Telugu inscriptions contain significant historical, cultural, political, religious, and geographical information. However, the information is generally present as unstructured text and is difficult to analyze automatically. After text extraction and translation, identifying the contextual meaning of the inscription remains a major challenge. Important entities such as kings, places, temples, dates, and dynasties must be identified accurately. In addition, relationships between people, locations, events, and historical periods need to be understood. Therefore, there is a need to develop an AI-based contextual interpretation and knowledge extraction module that can analyze translated inscription text, identify important historical entities, extract relevant facts and relationships, perform semantic analysis, generate meaningful summaries, and produce structured knowledge for further storage and analysis.

The engineering problem involves balancing several interacting requirements:

- Accurate identification of historical entities.
- Extraction of meaningful information from unstructured text.
- Understanding relationships between multiple entities.
- Generation of concise and contextually meaningful summaries.
- Reduction of incorrect or unsupported information generated by AI models.
- Production of structured output suitable for digital archiving and further analysis.

1.4 Project Aim

The aim of this project is to develop an **AI-powered contextual interpretation and knowledge extraction system for ancient Telugu inscriptions** using Natural Language Processing techniques and Large Language Models. The system aims to transform translated inscription text into meaningful and structured historical knowledge by identifying important entities, extracting relationships and facts, analysing semantic context, and generating concise summaries.

1.5 Project Objectives

1. To analyse translated ancient Telugu inscription text using Natural Language Processing techniques.
2. To identify important historical entities such as kings, places, temples, dynasties, dates, and events using Named Entity Recognition.
3. To extract relevant historical, cultural, political, and religious information from unstructured inscription text.
4. To identify relationships between people, locations, events, institutions, and historical periods using relation extraction techniques.
5. To perform semantic analysis for understanding the contextual meaning of the inscription.
6. To generate meaningful summaries that represent the major historical information contained in the inscription.
7. To convert extracted information into a structured format suitable for storage and further analysis.
8. To utilize Large Language Models for intelligent contextual interpretation and knowledge extraction.
9. To support digital preservation and easy retrieval of historical knowledge from ancient Telugu inscriptions.

1.6 Scope

1.6.1 Included Scope

The scope of this module includes:

- Analysis of translated ancient Telugu inscription text.
- Text preprocessing and normalization where required.
- Identification of important entities such as:
 - Kings and rulers
 - Places and geographical locations
 - Temples and religious institutions
 - Dates and historical periods
 - Dynasties
 - Historical events
- Information extraction from unstructured text.
- Relation extraction between identified entities.
- Semantic and contextual analysis.
- Generation of concise summaries.

1.6.2 Excluded Scope

The following activities are outside the primary scope of Module 3:

- Capturing inscription images.
- Advanced image enhancement and restoration.
- Optical Character Recognition implementation.
- Complete translation model development.
- Development of the complete digital archive frontend.
- Large-scale historical verification without expert support.
- Replacement of professional historians or epigraphists.

The project primarily receives the translated text as input and focuses on extracting contextual and structured information from it.

1.6.3 Assumptions

The proposed module operates under the following assumptions:

- The input text has already been extracted and translated by the preceding modules.
- The translated text contains sufficient information for contextual analysis.
- The LLM API is available during system execution.
- Extracted information can be validated or reviewed by experts when required.
- Historical interpretation generated by the system should be treated as AI-assisted analysis rather than final authoritative evidence.

1.6.4 Boundary Conditions

The performance of the system may be affected by:

- OCR errors present in the input text.
- Incomplete or damaged original inscriptions.
- Ambiguous historical names and locations.
- Limited context available in short inscriptions.
- Variations between ancient and modern Telugu language usage.

1.7 Expected Engineering Outcome

The expected outcome of the proposed project is an AI-based NLP module capable of transforming translated ancient Telugu inscription text into structured and meaningful historical knowledge.

The system is expected to provide the following outputs:

1. Entity Extraction

The system identifies important entities such as:

- Person or King
- Dynasty
- Place
- Temple
- Date
- Historical Period
- Event
- Religious or Cultural Institution

2. Information Extraction

The system extracts important facts from the inscription, including:

- Rulers mentioned
- Important locations
- Religious activities
- Donations
- Administrative actions
- Historical events
- Cultural information

3. Relation Extraction

The system identifies relationships such as:

King → ruled → Kingdom

King → donated to → Temple

Event → occurred at → Location

Person → belonged to → Dynasty

Temple → located in → Place

4. Contextual Interpretation

The system analyses the meaning of the inscription and provides an understandable interpretation of its historical significance.

5. Text Summarization

A The system analyzes the meaning of the inscription and provides an understandable interpretation of its historical significance.

6. Structured Knowledge Output

The structured output can later be integrated with a digital archive or interactive dashboard.

Entity Type	Extracted Entity	Context
Person	King Name	Mentioned as ruler
Place	Location Name	Place associated with event
Temple	Temple Name	Religious institution
Date	Historical Date	Time reference
Event	Donation	Historical activity

Table 1.1

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was conducted to understand the existing research and technologies related to ancient text restoration, Telugu Optical Character Recognition (OCR), Natural Language Processing, Large Language Models (LLMs), and contextual interpretation of historical documents.

The reviewed studies were selected to understand the strengths and limitations of existing approaches and to identify the research gap related to the contextual interpretation of ancient Telugu inscriptions.

The review particularly focuses on whether existing approaches provide an end-to-end mechanism for understanding historical context, identifying entities, extracting relationships, and generating structured knowledge from ancient inscription text.

2.2 Review of Existing Work

The literature review was conducted to understand the existing research and technologies related to ancient text restoration, Telugu Optical Character Recognition (OCR), Natural Language Processing, Large Language Models (LLMs), and contextual interpretation of historical documents.

Relevant studies were identified based on the following areas:

- Ancient text restoration using Artificial Intelligence.
- Deep learning techniques for Telugu character recognition.
- OCR methods for Indian languages.
- Large Language Models for translation of ancient languages.
- Named Entity Recognition and Information Extraction.
- Contextual interpretation and knowledge extraction from historical text.

2.2.1 Deep Neural Networks for Ancient Text Restoration

Assael et al. (2022) presented a deep learning approach for restoring and attributing ancient texts using neural networks. The system, known as Ithaca, demonstrated the potential of Artificial Intelligence in restoring damaged portions of ancient Greek inscriptions and predicting their geographical and chronological information.

2.2.2 Deep Learning-Based Telugu OCR

Dhanikonda et al. (2022) proposed an efficient deep learning model for Telugu Optical Character Recognition. The research addressed the complexity of Telugu characters using deep learning, segmentation, and tagging techniques.

Telugu script contains complex character structures, vowel modifiers, and combinations that create challenges for traditional OCR systems. The proposed approach demonstrated the importance of deep learning in improving Telugu character recognition.

2.2.3 Deep Learning-Based Telugu OCR Studies

Vijaya Saradhi et al. (2023) presented a comprehensive study of deep learning-based Telugu OCR techniques. The study reviewed CNN-based approaches, OCR methods, and text segmentation techniques used for Telugu language recognition.

The research highlighted the importance of OCR in digitizing Indian-language documents and preserving historical records. It also discussed the role of deep learning in improving recognition performance.

2.2.4 Large Language Models for Ancient Text Translation

Wannaz and Miyagawa (2024) evaluated Large Language Models for translating Coptic and Ancient Greek ostraca. The study examined the ability of multiple LLMs to translate historical and ancient-language texts.

The research demonstrated that Large Language Models can assist in the translation and interpretation of ancient texts. LLMs have the ability to consider linguistic context and generate fluent translations.

However, the study also highlighted an important limitation: LLMs may generate incorrect or unsupported information, especially when working with low-resource languages or incomplete historical material.

2.2.5 Instruction-Tuned Language Models for Ancient Text Restoration

Cullhed (2026) explored instruction-tuning pretrained causal language models for restoring ancient Greek papyri and inscriptions. The work investigated the use of a pretrained LLM, including Llama-based approaches, for restoring missing characters and portions of ancient texts.

2.2.6 Ancient Character Recognition and Cultural Heritage Preservation

Several recent studies have applied deep learning and computer vision techniques to ancient manuscripts, inscriptions, and historical documents.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Ancient Text Restoration using Deep Neural Networks	Helps restore damaged ancient texts and predict historical information	Mainly focused on Ancient Greek and not Telugu inscriptions	Demonstrates the use of AI for ancient text analysis
Deep Learning-Based Telugu OCR	Improves recognition of Telugu characters	Focuses mainly on text recognition and not contextual interpretation	Provides the textual input required for further NLP analysis
LLM-Based Ancient Text Translation	Provides context-aware and fluent translation	May generate inaccurate information and requires validation	Supports contextual understanding of historical text

Table 1.2

2.4 Research / Engineering Gap

Existing research has made significant progress in ancient text restoration, Telugu OCR, character recognition, and LLM-based translation. However, most existing approaches focus on individual tasks rather than providing an integrated solution for ancient Telugu inscriptions. OCR systems can extract text but do not provide a complete understanding of its historical context. Similarly, translation systems may convert ancient text into modern languages but do not automatically identify important entities such as kings, places, temples, dates, and historical events or determine the relationships between them. There is also limited research focused specifically on the contextual interpretation and structured knowledge extraction of ancient Telugu inscriptions. Therefore, there is a need for an integrated NLP-based system that combines Named Entity Recognition, Information Extraction, Relation Extraction, Semantic Analysis, and Text Summarization to transform unstructured inscription text into meaningful and structured historical knowledge.

2.5 Proposed Contribution

The proposed project contributes to the development of an AI-powered system for the contextual interpretation and knowledge extraction of ancient Telugu inscriptions. Unlike existing approaches that mainly focus on individual tasks such as text recognition, restoration, or translation, the proposed Module 3 integrates multiple NLP techniques to analyze the meaning and historical context of the translated inscription text. The system identifies important entities such as kings, places, temples, dates, and historical events using Named Entity Recognition and extracts relevant historical, cultural, and religious information through Information Extraction. It further identifies relationship.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The proposed system is intended to support researchers, historians, students, and general users who are interested in understanding ancient Telugu inscriptions. The system should provide meaningful contextual interpretation and structured historical information from the translated inscription text.

Stakeholder / User Requirements

Stakeholder	Requirement
Researchers and Historians	Extract important historical entities and relationships from inscriptions
Students	Obtain simple summaries and understandable historical information
General Users	Access structured and easy-to-understand inscription content
System Administrator	Store and manage extracted information efficiently

Table 3.1

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Identify historical entities such as kings, places, temples, and dates	Functional	Extract relevant entities from input text
Extract important historical information	Functional	Generate structured facts from inscription text
Identify relationships between entities	Functional	Produce meaningful entity relationships
Generate contextual summaries	Functional	Generate a concise summary for each inscription

Table 3.2

Requirement	Type	Measurable Target
Accuracy	Non-Functional	Generate contextually relevant output
Performance	Non-Functional	Process text within an acceptable response time
Usability	Non-Functional	Produce understandable and structured output
Reliability	Non-Functional	Handle incomplete or unclear input appropriately

Table 3.3

3.2 Constraints & Applicable Standards

The proposed system operates under certain technical and ethical constraints. The quality of the output depends on the quality of the translated input text. OCR errors or translation errors from previous modules may affect entity identification and contextual interpretation.

Standard / Guideline	Requirement	How Applied in This Project
Data Privacy Principles	Protect data and avoid unnecessary personal data collection	The module processes inscription text and does not require personal user information
Responsible AI Principles	Reduce inaccurate and misleading AI output	Output is treated as AI-assisted interpretation and can be reviewed by experts
Academic Integrity Guidelines	Acknowledge sources, datasets, tools, and AI assistance	Relevant resources and tools are documented in the report
Software Documentation Practices	Maintain clear implementation and testing records	System workflow, tools, testing, and results are documented

Table 3.4

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Input	Translated ancient Telugu inscription text	Text	High
Entity Extraction	Identify important historical entities	Extracted entities	High
Information Extraction	Extract historical and cultural facts	Structured information	High
Relation Extraction	Identify relationships between entities	Relations	High
Summarization	Generate concise contextual summary	Text summary	Medium

Table 3.5

3.4 Alternative Solutions & Evaluation

This approach uses predefined rules, keywords, and patterns to identify entities and extract information. It is simple and easy to implement but may not handle complex sentence structures or different historical contexts effectively.

Alternative 1: This approach uses trained machine learning models for entity recognition and classification. It can provide better automation than rule-based methods but requires a sufficiently large labelled dataset, which may be difficult to obtain for ancient Telugu inscriptions.

Alternative 2: This approach uses Large Language Models to understand the context of the translated inscription text and extract entities, facts, relationships, and summaries. It provides greater flexibility but requires careful validation because LLMs may generate inaccurate information.

Criterion	Weight	Rule-Based NLP	Traditional ML	LLM-Based Approach
Contextual Understanding	30%	2	3	5
Flexibility	25%	2	3	5
Dataset Requirement	20%	5	2	4
Implementation Feasibility	25%	5	3	4
Total Score	100%	3.35	2.80	4.55

Rating Scale: 1 – Very Low, 2 – Low, 3 – Moderate, 4 – High, 5 – Very High

Based on the evaluation, the **LLM-Based Contextual Interpretation approach** was selected because it provides better contextual understanding and flexibility for analysing historical text.

3.5 Selected Design & Detailed Design

The input to the system is translated inscription text received from the previous translation module. The text is first preprocessed to remove unnecessary characters and normalize the input. The processed text is then analyzed using the LLM-based knowledge extraction pipeline.

The input to the system is translated inscription text received from the previous translation module. The text is first pre-processed to remove unnecessary characters and normalize the input. The processed text is then analysed using the LLM-based knowledge extraction pipeline.

The system performs the following operations:

1. Text preprocessing and normalization.
2. Named Entity Recognition.
3. Information Extraction.
4. Relation Extraction.
5. Semantic Analysis.
6. Text Summarization.

Flow Chart

Translated Inscription Text



Text Preprocessing & Normalization



Named Entity Recognition (NER)



Information Extraction



Relation Extraction



Semantic Analysis



Text Summarization

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
LLM-Based Analysis	Rule-Based NLP	Better contextual understanding	Possible inaccurate output	Selected because historical text requires contextual interpretation
Structured Output	Plain Text Output	Easy storage and retrieval	Requires additional processing	Selected for digital archiving and future analysis
General LLM	Custom-Trained Model	Easy implementation and flexible analysis	Less domain-specific	Selected due to limited availability of labelled ancient Telugu datasets
Automated Analysis	Fully Manual Analysis	Faster processing of large text collections	Requires expert validation	Selected as an AI-assisted approach

3.7 Sustainability, Ethics, Safety, Risk & Security

This is the section institutional guidelines flag most for vague statements. Never write 'sustainability was considered' — the table must show what evidence was evaluated and how it changed the design decision.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Digital preservation	Historical information can be stored digitally	Structured output was selected
Ethics	AI hallucination	LLMs may generate inaccurate information	Expert validation is recommended
Health & Safety	Software-based development	No physical interaction with hazardous systems	Standard computer safety is sufficient
Security	Data and API protection	API keys can be exposed if improperly managed	Secure handling of credentials is required

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect entity extraction	Medium	Medium	Medium	Review and validation of extracted entities	Low
LLM hallucination	Medium	High	High	Use structured prompts and expert verification	Medium
Poor input quality	High	Medium	High	Apply input cleaning and error handling	Medium
Incorrect relationship extraction	Medium	Medium	Medium	Validate extracted relationships	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The proposed Module 3 was developed using an NLP-based approach for contextual interpretation and knowledge extraction from translated ancient Telugu inscription text. The development process begins by receiving the translated text from the previous module. The input text is cleaned and prepared for further analysis. An LLM-based approach is then used to identify important historical entities, extract relevant information and relationships, understand the semantic context, and generate a concise summary. Finally, the extracted information is organized into a structured format for storage and further analysis.

Resource	Purpose
Translated Inscription Text	Input for contextual analysis
Python	Implementation of the NLP processing workflow
Gemini API	LLM-based contextual interpretation and knowledge extraction
JSON	Structured representation of extracted information

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Use subheadings (not chapters) for whichever of hardware/software/fabrication apply. Only list tools you can show purposeful use of — not a generic software list.

4.2.1 Input Processing

The system accepts translated ancient Telugu inscription text as the input. The text is prepared for analysis by removing unnecessary formatting and ensuring that the input is in a suitable form for processing.

4.2.2 Contextual Interpretation

The processed text is provided to the Large Language Model through the Gemini API. The LLM analyzes the content and identifies the overall meaning and historical context of the inscription.

The prompt is designed to request structured and relevant information rather than unrestricted text generation.

4.2.3 Named Entity Recognition and Information Extraction

The system identifies important entities and information from the text. The main categories include:

- Kings and rulers
- Places
- Temples
- Dates
- Dynasties

- Historical events
- Cultural information

4.2.4 Relation Extraction

After identifying the important entities, the system extracts relationships between them.

Examples include:

- King → ruled → Region
- King → donated to → Temple
- Temple → located in → Place
- Event → occurred at → Location

This process helps convert individual pieces of information into connected historical knowledge.

4.2.5 Text Summarization

The system generates a concise summary of the inscription. The summary focuses on the major people, places, events, and activities mentioned in the text.

This allows users to understand the main historical significance of an inscription without reading the complete text repeatedly.

Tool / Software	Purpose	Stage Used
Python	Core implementation	Development
Gemini API	Contextual interpretation	NLP Processing
JSON	Structured output generation	Knowledge Storage
Jupyter Notebook / Colab	Development and testing	Testing
LLM Prompts	Controlled information extraction	Analysis

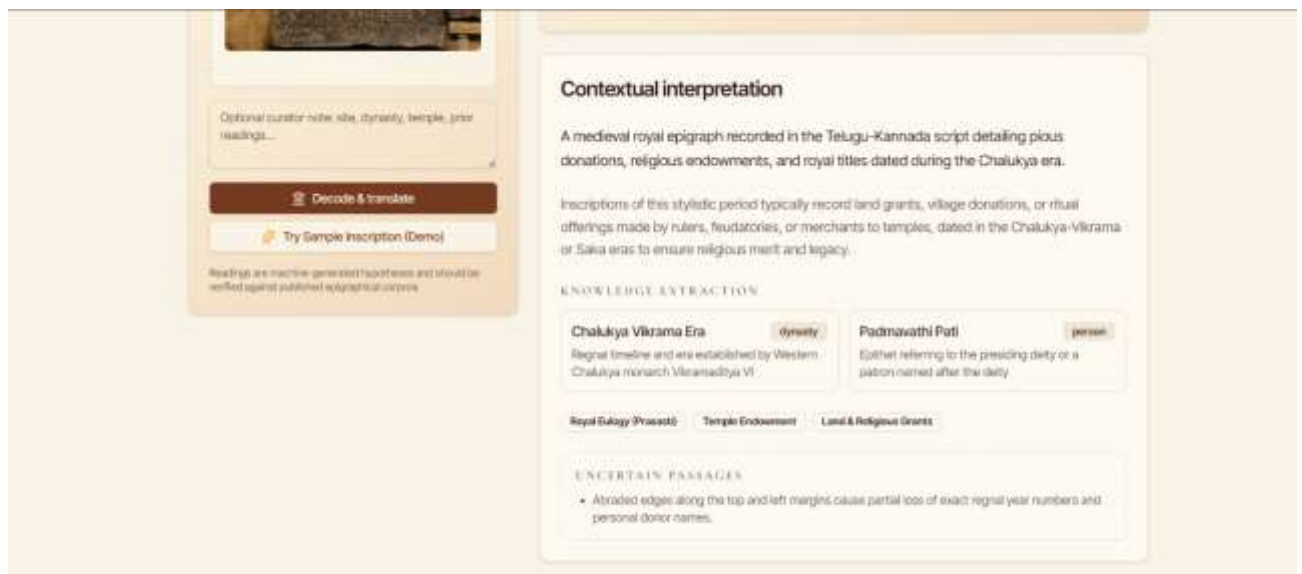


Fig 4.1

4.3 Test Plan & Verification

The testing process was designed to verify whether the proposed system performs the required contextual interpretation and knowledge extraction tasks.

The module was tested using translated inscription text containing different types of historical information.

Requirement	Test Method	Acceptance Criterion
Entity Extraction	Provide text containing known entities	Important entities should be identified
Information Extraction	Analyze historical statements	Relevant facts should be extracted
Relation Extraction	Provide connected historical information	Meaningful relationships should be generated
Summarization	Provide complete inscription text	Concise summary should retain key information
Requirement	Test Method	Acceptance Criterion

Specification	Required Value	Achieved Value	Status
Entity Identification	Identify key entities	Entities categorized successfully	Pass
Information Extraction	Extract historical facts	Relevant facts generated	Pass
Relation Extraction	Identify entity connections	Relationships generated	Pass
Summarization	Generate concise output	Meaningful summary generated	Pass

4.4 Results

Present measurements, graphs, and key data. Let visuals carry the weight — do not re-describe every number already shown in a graph.

The implementation demonstrates that an LLM-based approach can assist in analysing translated ancient Telugu inscription text and converting it into meaningful information.

The system produced the following outputs:

Entity Identification

Important historical entities such as people, places, temples, dates, and events can be identified from the input text.

Information Extraction

The system can extract relevant historical and cultural information embedded within unstructured text.

Relation Extraction

Relationships between entities can be represented in a structured form, improving the understanding of the historical context.

Text Summarization

The module generates a concise summary representing the major content and significance of the inscription.

4.5 Data Analysis & Engineering Judgment

Interpret the data (statistical/error/accuracy/performance metrics as relevant). Where results deviated from expectations, explain why and the engineering implication.

The results indicate that the LLM-based approach is useful for analysing unstructured historical text because it can consider the context surrounding individual words and sentences.

Traditional keyword-based approaches may identify a name or location but may not understand the relationship between them. In comparison, the proposed approach attempts to interpret the meaning of the complete statement.

However, the quality of the output depends on several factors:

- Quality of OCR output from the previous module.
- Accuracy of the translated text.
- Amount of historical context available.
- Clarity of the inscription content.
- Reliability of the LLM response.

An important engineering consideration is the possibility of hallucination. Large Language Models may generate information that is not explicitly supported by the input text. Therefore, structured prompting and human validation are important when the output is used for historical research.

The engineering judgment from the implementation is that LLMs should be used as AI-assisted interpretation tools rather than as completely independent historical authorities.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Analyze translated inscription text	NLP-based processing workflow implemented	Fully Achieved
Identify important historical entities	Entity extraction output generated	Fully Achieved
Extract historical information	Structured facts generated	Fully Achieved
Identify relationships	Entity relationships generated	Fully Achieved
Generate summaries	Contextual summary produced	Fully Achieved

Table 4 .1

4.7 Strengths & Limitations

State limitations honestly — do not claim the system is perfect.

Strengths:

- Integrates multiple NLP tasks into a single workflow.
- Provides contextual interpretation instead of simple keyword matching.
- Identifies important historical entities.
- Extracts relationships between people, places, and events.
- Generates concise summaries.
- Produces structured information suitable for digital archiving.
- Can reduce manual effort in analysing large collections of text.

Limitations

- Output quality depends on the accuracy of the input text.
- OCR and translation errors can affect contextual interpretation.
- LLMs may generate inaccurate or unsupported information.
- Limited availability of specialized ancient Telugu datasets.
- Expert validation may be required for important historical conclusions.
- The system may have difficulty interpreting incomplete or highly damaged inscriptions.

4.8 Project Planning, Roles & Budget

Gantt chart or milestone table, roles/contribution table, and a brief cost table — this replaces the old standalone Project Management chapter.

[Insert Gantt chart / milestone timeline here.]

Phase	Activity	Status
Phase 1	Requirement analysis	Completed
Phase 2	Literature review	Completed
Phase 3	System design	Completed
Phase 4	Module implementation	Completed
Phase 5	Testing and result analysis	Completed

Table 4.2

Student	Assigned Responsibility	Actual Contribution
B. Akshaya	Module 3 – Contextual Interpretation and Knowledge Extraction	Implemented entity extraction, information extraction, relation extraction, semantic analysis, summarization, and structured knowledge generation

Table 4.3

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The proposed Module 3 focused on the contextual interpretation and knowledge extraction of translated ancient Telugu inscription text using Natural Language Processing techniques and Large Language Models. Ancient inscriptions contain valuable historical, cultural, religious, political, and geographical information, but understanding this information manually can be difficult and time-consuming. The proposed system addresses this challenge by applying multiple NLP techniques, including Named Entity Recognition, Information Extraction, Relation Extraction, Semantic Analysis, Text Summarization, and Knowledge Extraction. The system identifies important historical entities such as kings, places, temples, dates, dynasties, and events. It also extracts meaningful facts and relationships between these entities to provide a better understanding of the historical context.

The implementation demonstrates that an LLM-based approach can assist in transforming unstructured translated text into meaningful and structured historical knowledge. The generated summaries and structured outputs make the information easier to understand, store, search, and analyse. The module also supports the broader objective of preserving ancient Telugu cultural heritage digitally. By converting inscription information into structured knowledge, the system can assist researchers, students, and historians in accessing historical information more efficiently.

5.2 Future Work

The proposed system can be further improved in several ways. A larger and well-annotated dataset of ancient Telugu inscriptions can be developed to improve the accuracy of entity recognition and contextual interpretation. Future work can also focus on creating specialized NLP models trained specifically on ancient Telugu vocabulary, historical terminology, and inscription styles. The system can be enhanced by incorporating a knowledge graph to represent complex relationships between kings, dynasties, places, temples, and historical events. This would allow users to visually explore historical connections. Another important improvement is the inclusion of a human-in-the-loop validation mechanism, where historians or domain experts can review and correct the AI-generated results. Such validated data can later be used to improve the performance of the system. The module can also be integrated with the digital archive and interactive dashboard to provide semantic search and question-answering capabilities. Users could search for inscriptions based on a king, dynasty, location, temple, or historical period.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

Through the development of Module 3, the following new knowledge was acquired:

- Understanding of Natural Language Processing techniques for historical text analysis.
- Knowledge of Named Entity Recognition and Information Extraction.
- Understanding of Relation Extraction between different historical entities.
- Experience in using Large Language Models for contextual interpretation.
- Knowledge of semantic analysis and text summarization.
- Understanding of structured knowledge representation using JSON and tables.
- Awareness of AI hallucination and the importance of validating AI-generated information.

Engineering & Professional Skills Developed:

The project contributed to the development of the following skills:

- Natural Language Processing implementation.
- Python programming.
- Prompt design for Large Language Models.
- Text analysis and information extraction.
- Problem-solving and system design.
- Testing and evaluation of AI-generated outputs.
- Technical documentation and report writing.
- Understanding of ethical and responsible AI practices.

Individual Reflection (each student, 4–5 lines):

A key engineering decision was selecting an LLM-based approach instead of relying only on rule-based methods. This decision was made because historical text requires contextual understanding, and the available specialized datasets for ancient Telugu inscriptions are limited. The LLM-based approach provided greater flexibility in identifying entities, relationships, and important historical information.

During this work, I independently learned about Named Entity Recognition, Information Extraction, Relation Extraction, Semantic Analysis, and structured knowledge generation. I also gained an understanding of the limitations of Large Language Models, particularly the possibility of generating inaccurate information.

REFERENCES

- [1] Y. Assael *et al.*, “Restoring and Attributing Ancient Texts Using Deep Neural Networks,” *Nature*, vol. 603, pp. 280–283, 2022.
- [2] S. R. Dhanikonda, S. Ponnuru, M. L. Ramanaiah, *et al.*, “An Efficient Deep Learning Model with Interrelated Tagging Prototype with Segmentation for Telugu Optical Character Recognition,” *Scientific Programming*, 2022.
- [3] V. K. Sonthi, S. Nagarajan, and N. Krishnaraj, “Deep Learning Based Residual Network Features for Telugu Printed Character Recognition,” *Intelligent Automation & Soft Computing*, vol. 34, no. 3, pp. 1725–1736, 2022.
- [4] G. Suresh, C. R. Prasad, and S. Kollem, “Telugu Optical Character Recognition Using Deep Learning,” in *Proceedings of the 3rd International Conference for Emerging Technology (INCET)*, 2022.
- [5] B. Nair, K. V. Raj, K. Mangalath, V. Pai, and E. V. Sreejil, “Ancient Epic Manuscript Binarization and Classification Using False Color Spectralization and VGG-16 Model,” *Procedia Computer Science*, vol. 218, pp. 631–643, 2023.
- [6] R. Priya, S. Karthikeyan, J. Indra, *et al.*, “Self-Adaptive Hybridized Lion Optimization Algorithm With Transfer Learning for Ancient Tamil Character Recognition in Stone Inscriptions,” *IEEE Access*, 2023.
- [7] M. Wannaz and S. Miyagawa, “Assessing Large Language Models in Translating Coptic and Ancient Greek Ostraca,” *ACL Anthology*, 2024.
- [8] N. Cullhed, “Instruction-Tuning Pretrained Causal Language Models to Restore Ancient Greek Papyri and Inscriptions,” 2026.
- [9] Google, “Gemini API Documentation,” Google AI for Developers, accessed 2026.
- [10] Python Software Foundation, “Python Documentation,” Python Software Foundation, accessed 2026.

APPENDICES

Appendix A – Detailed Calculations

The proposed Module 3 is primarily a software and NLP-based system. Therefore, no major mathematical or hardware engineering calculations are required. However, the performance of the system can be evaluated using appropriate metrics in future implementations.

For example:

Precision

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall

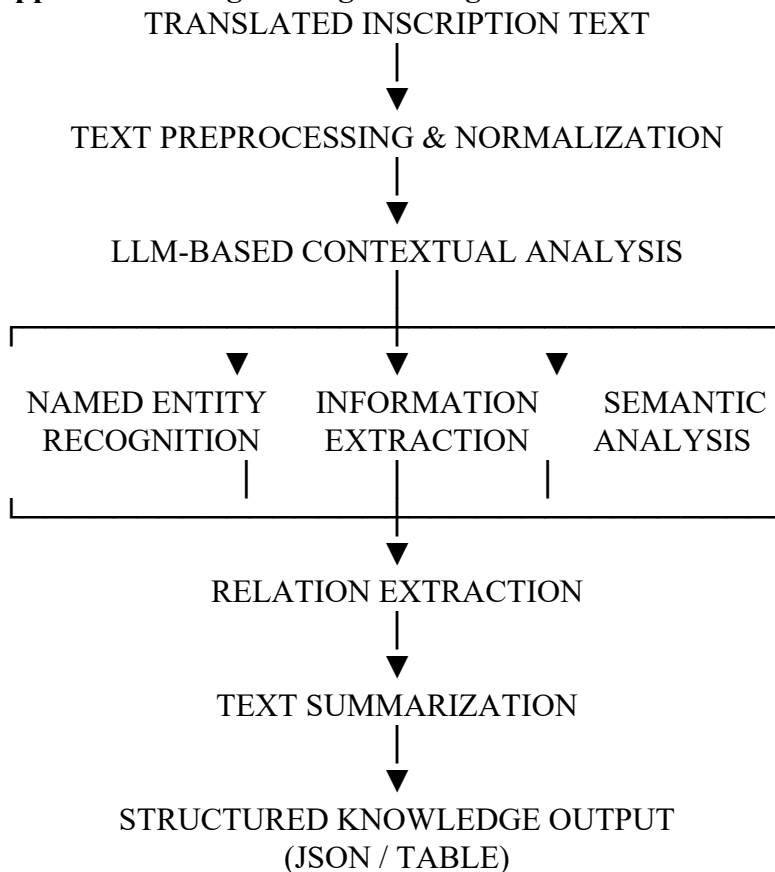
$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1-Score

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics can be used to evaluate the performance of entity extraction and information extraction when a labelled ground-truth dataset is available.

Appendix B – Engineering Drawings / Schematics



Appendix C – Source Code / Code Repository Information

Only essential code excerpts should normally be included in the report. Complete source code may be maintained separately in the approved repository.

Code:

```
def analyze_inscription(text):  
    prompt = f"""  
    Analyze the following translated ancient Telugu inscription.
```

Extract the following information:

1. Persons or Kings
2. Places
3. Temples or Institutions
4. Dates or Historical Periods
5. Historical Events
6. Important Relationships
7. Historical Summary

Return the output in a structured format.

Inscription Text:

```
{text}  
"""
```

```
response = generate_llm_response(prompt)
```

```
return response
```

Appendix D – Datasheets

The proposed Module 3 is a software-based NLP system and does not use hardware components requiring technical datasheets.

Software / Tool	Purpose
Python	Core programming language
Gemini API	LLM-based contextual interpretation
Jupyter Notebook	Development and testing
Google Colab	Cloud-based development environment
JSON	Structured data representation

Appendix E – Experimental / Raw Data

The inscription records that King X made a land donation to Temple Y located in Place Z during a particular period.

Category	Extracted Information
Person	King X
Temple	Temple Y
Place	Place Z
Event	Land Donation
Relationship	King X → donated to → Temple Y
Relationship	Temple Y → located in → Place Z

The inscription records a land donation made by King X to Temple Y located in Place Z.

Note: Actual experimental inputs and outputs from your implementation should be added here as screenshots or tables if available.

Appendix F – Ethics / Institutional Approval

The proposed Module 3 is developed for academic and cultural heritage research purposes.

The following ethical considerations were followed:

- Historical information should not be intentionally fabricated.
- AI-generated outputs should be treated as machine-assisted interpretations.
- Important historical conclusions should be validated by domain experts.
- Sources, datasets, software, and AI-assisted tools should be appropriately acknowledged.
- API credentials and system access information should be protected.
- The system should respect the cultural significance of ancient Telugu heritage.

Appendix G – Project Meeting / Progress Records

Phase	Activity	Status
Week 1	Understanding Module 3 requirements	Completed
Week 2	Literature review	Completed
Week 3	NLP technique analysis	Completed
Week 4	System design	Completed
Week 5	LLM-based implementation	Completed
Week 6	Testing and result analysis	Completed
Week 7	Report preparation	Completed

Appendix H – User/Industry Feedback

The proposed system is intended for use by researchers, historians, students, and individuals interested in Telugu cultural heritage.

Users can evaluate the system based on:

- Relevance of extracted entities.
- Accuracy of historical information.
- Quality of generated summaries.

Appendix I – Publications / Patents / Competitions / Product Outcomes

At present, no publication, patent, or competition outcome is claimed for the proposed Module 3.

Status: Not Applicable / To be updated if applicable.

Future work may include:

- Research paper publication.
- Development of a complete digital archive.
- Integration with an interactive dashboard.
- Development of a specialized ancient Telugu NLP model.

Appendix J – Individual Contribution Evidence

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution
B. Pallavi (192424243)	Module 1 – Image Processing and Text Extraction	Image preprocessing, noise removal and OCR workflow	Tested image enhancement and text extraction results	Module 1 documentation	25%
A. Ashriktha Reddy (192424172)	Module 2 – Translation Using LLMs	Text preprocessing, OCR error correction and LLM-based translation	Tested translation quality and contextual relevance	Module 2 documentation	25%
B. Akshaya (192424336)	Module 3 – Contextual Interpretation and Knowledge Extraction	NER, Information Extraction, Relation Extraction, Semantic Analysis and Summarization	Tested extracted entities, relationships and summaries	Module 3 documentation	25%
Ch. Haritha (192425280)	Module 4 – Digital Archive and Interactive Dashboard	Archive design, frontend/backend integration, search and dashboard development	Tested search, storage and dashboard functionality	Module 4 documentation	25%
Total					100%

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location
Problem identification and NLP solution	SO1	Ch. 1, Ch. 2
Requirements, design and constraints	SO2	Ch. 3
Report, diagrams and presentation	SO3	Whole Report + Viva
AI ethics and security	SO4	Ch. 3.7
Team roles and planning	SO5	Ch. 4.8
Testing, results and analysis	SO6	Ch. 4.3–4.6
Learning NLP and LLM techniques	SO7	Ch. 5.3
Software and security practices	Standards	Ch. 3.2
Digital preservation of inscriptions	Sustainability	Ch. 3.7
Technical risks and mitigation	Risk Assessment	Ch. 3.7
API and data security	Security	Ch. 3.7
Integration of NLP modules	System Engineering	Ch. 3.5
Milestones and task allocation	Project Management	Ch. 4.8
Individual module contributions	Individual Contribution	Contribution Statement